Introduction to the Devops and Cloud Computing learning program

- Think of this course as a legos building block, we will connect multiple pieces together to build something strong and complete
- Till now we have learnt Linux we will learn some other core skills as well such as Virtualization, networking, scripting
- You can assume this structure to be something like chassis of a car, Once it's ready you can start fitting engine, the wheels, the body
- In Devops and cloud computing world it means AWS, Jenkins, GitHub, Terraform and many other devops tools which we will discuss very soon.
- Now as a car can't exist without it's chassis similarly don't underestimate the power of the basics which we are going to discuss for the first few weeks and later based on these we will scale and learn more advanced topics like AWS, Docker, Maven, Jenkins, Nexus, SonarQube, Terraform, Kubernetes etc.

Virtualization

- If we have to build a lab connecting 10 to 15 computers in today's time it would be really expensive, but using virtualization you could easily build as many virtual users/systems you want and can create an entire virtual machine lab inside your computer.
- Virtual machine is simply a computing system inside your system, while it's not an actual one but this virtual system can have operating systems could be connected together.
- Before virtualization one computer having multiple OS's
- To run apps/websites/services we need servers(huge powerful physical computers in data centers)
- For every service a separate server is required for the sake of isolation as if all the services would be running on just one system because of just one service failing the whole server would crash and everything would come to a pause
- In server architecture, **isolation** means running different services on **separate servers or environments** so that problems in one service do not affect others. This improves **security, performance, reliability, and scalability**.
- In a web application, different services run on separate servers to avoid interference. The **web server** (using Nginx) handles user requests from the internet. The **application server** (running Node.js or Spring Boot) processes the business logic. The **database server** (using MySQL) stores and manages data. Sometimes a **cache server** like Redis is also used to speed up

responses. By isolating these services on different servers, failures or heavy load in one service do not affect the others.
- Now when VMware came into the game with virtualization all this could be done on just one system, as VMware allows Virtualization.
- Although we are talking about server virtualization but there are various other type of virtualization as well such as network virtualization, storage virtualization
- On top of computer hardware we have a software tool called hypervisor on which you could create Virtual machines each having their own os and each running independent of other

Terminologies

- Host OS: The operating system on your actual machine
- Guest OS: The operating system on your virtual machines
- VM: Acronym for virtual machine
- Snapshot: It's a way of taking backup of the virtual machine, as virtual machines in reality are just a set of files so they could be backed up very easily
- Hypervisor: It is the tool which let's us create virtual machine, it enables virtualization
    1. **Type 1(Bare metal):** It's only for production it won't let you use your computer for other purposes,,and it runs directly on the physical computer like an os . **ex VMware ESXi or Zen Hypervisor**
    2. **Type 2(Testing and study purpose):**It runs as a software which you can install on any computer. It's just for learning and testing purpose. In this pep course we are not going to run production. **Ex oracle VM VirtualBox, VMware server.** In our learning we will be using VM VirtualBox

Creating Virtual Machines

- **Centos** and **Ubuntu** are two very popular flavor of Linux OS but there are various other options as well.
- There are two methods to create VM's
- **Method 1:** First is manual which is wizard based where we select multiple options to create VM Box
- **Method 2:** The second is automated you create a text file and then issue a command and VM Box will be created.
- Even though automated option is available it's always better to first know how to manually do it first.
- Later we will migrate to AWS so we'll be doing everything on cloud computing but for initial practice you need Linux VM's.

- For windows we will use **Oracle VM Virtualbox** as Hypervisor but if you are using **macOS m1 and m2** you will need **ISO file CentOS** and **Ubuntu.** These ISO files are basically installation files which we are going to attach to our VM.
- We will need a login tool **Git bash** or **Putty**.

Method 1: Creating Virtualization manually

Prerequisites

- If you are a mac user you don't need to do anything
- But for window user follow these steps to have all the prerequisites sorted:
    1. Enable Virtualization in BIOS, It would be one of the following options(VTx, Secure virtual machine, Virtualization) in the BIOS settings. Restart your computer and press one of F2, F10, F12, Del or maybe ESC depending on what pc you are using and which option leads to the BIOS settings for that pc.
    2. Go to start menu > Turn windows features on or off > Disable all the undermentioned settings in the list of options
        - Microsoft Hyperv
        - Windows Hypervisor platform
        - Windows Subsystem for linux
        - Docker Desktop
        - Virtual Machine Platform
    3. Reboot your computer

Process for creating a vm

Step1. Open **Oracle Virtual Box Manager**

Step2. Click on the **new** option indicated by the gear sign

Step3. Enter the name for your new virtual machine in the first column of the menu, name could be anything you prefer.

Step4. In **type** the OS type will be entered we will keep it **Linux**

Step5. Select **Subtype** according to what type of Virtual machine you are creating for example if you are creating **centos vm** then it would be **Red Hat** and **version** would be **Red Hat(64 bit)**

Step6. If **version** is **Red Hat(64-bit)** totally fine otherwise if it is **32-bit** that means virtualization is disabled check in BIOS settings and enable virtualization following the process shared earlier.

Step7. In **hardware** select **base memory** and **processors/CPU's** you could select base memory and CPU's according to your requirement/choice

Step8. **Virtual hard disk** assign whatever size you want to assign to the hard disk for this virtual computer, then if you want to pre allocate the whole size you can check the respective option else the space will be dynamically allocated as per the use only and over the time depending upon the requirement it will be reallocated. Generally it's better to keep it dynamic so for that keep the pre allocate option unchecked

Step9. Now the system is ready but it don't have any **os** installed on it. So depending on which system you are creating you have to install the os for that.

Now your VM is ready next step is providing the booting fill and connecting to the network

Adding centOS file in VM similarly you can add files for other subtypes as well

- Search for Cetos stream iso download ISO will be an image file which will attach to our VM and will power on our VM and our VM will start installation from this image file
- Select Index of /9-stream/BaseOS/x86_64/iso
- Download boot.iso file
- Then come back to **Oracle Virtual Box Manager** and open **settings** for your **vm**,go to **expert**, select **storage**. In **controller IDE** click on **empty** select the downloaded **centos file**.Click over **dropdown** option at extreme right and click on **choose a Disk File** then choose your downloaded **centos file** and then **check the Live CD/DVD** checkbox
- So here we are attaching this ISO file so that when we power on the VM it starts booting from this file

Connecting VM with network adapter

- Now for connecting your virtual machine with internet we need to connect it to the wifi adapter. Then router will give an IP address to this VM network adapter assuming it's just a normal network adapter. For connecting vm to wifi adapter go to **network settings** for your **vm**, keep the first adapter as it is it's default in virtual box, **select Adapter 2(select Enable Network Adapter, in Attached to drop down menu select Bridged network)**
- Go to **settings**, select **System settings**, open **motherboard settings** and in **pointing devices** select **USB tablet**, it will be really helpful later for using mouse for cursor for this vm.

Note: For checking your computer's network adapter and it's IP address go to terminal and enter command for **windows: ipconfig** and for **mac: ifconfig.** Here we need only two imp things one is **IP address** of the **network adapter for this device** and **IP address** of the **router** that is of **Default Gateway.**

Power on the VM

Now we will power on our VM and the whole booting process for the virtual machine will start

- Once you follow the above steps VM is created so we can power it on by selecting it and clicking on start.
- Now click on the blank screen
- It's going to capture your mouse cursor for which it will ask you for a permission so select capture
- We've to wait for some time it's going to check all the hardware and everything of this system then it will start the installation in some time. It will ask you for the **language** select **English**
- Then on next screen first we need to select **Installation Destination** that will be the virtual hard disk of this VM, select **virtual hard disk 20 gb** and click on done. Then select **Installation Destination** once again select **virtual hard disk 20 gb Storage configuration Automatic** and click on done again. In future we will discuss linux partitioning and it's file system structure for now just select automatic partitioning.
- Next click on **Network & Host Name** you will see two network adapters here first one is the **NAT adapter** and then second one is the **bridged adapter.** You can verify that by selecting the network adapters and checking their IP addresses. Select the second network adapter then give some **host name** whatever you prefer, I am giving the same name as VM machine name but you can give some other name as well. Click done.
- Next step is **setting up root password**, root is the **admin user** in Linux we will discuss it in more detail later when we discuss linux. For now just set the password.
- Click on begin installation

After Installation

- After installation is completed it will ask you for reboot. You could either reboot it.
- Or go to Oracle Virtual Box manager We are going to **power off VM**, so for it **right click** on your **os** select **stop** choose **ACPI shut down**.
- Once it's completely powered off go to **settings > storage > select iso > remove disk from Virtual Drive(on right side there would be a cd like sign for it).** We have to remove the ISO disk from virtual drive it's like removing a cd from the cd rom so that it doesn't boots again with ISO Cd because that way it will restart the whole installation process again and everytime when the system would be restarted.
- After this you could start the VM it's ready for use just select the VM and click on start and you will be able to use it